

Concurrency and Distributed Systems

Week 9 – Part 3: Go and Erlang Contrast

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Part 3 Roadmap

1 Language Models

2 Tradeoffs

3 Demo

Why Compare Languages?

Course point

A runtime model changes what "concurrency" costs and what it makes easy.

- Threads are not the only abstraction.
- Shared memory is not the only communication model.
- Fault handling can be part of the concurrency model.

Python, Go, Erlang

Runtime	Main idea	Communication
Python	GIL constrains CPU-bound threads	shared objects, queues
Go	goroutines scheduled by runtime	channels
Erlang	lightweight actors/processes	message passing

- Goroutines are cheap compared with OS threads.
- The runtime multiplexes goroutines onto OS threads.
- Channels encourage communication by passing values.

- Actors communicate through messages.
- State is isolated per process.
- Supervision and failure recovery are central ideas.

Run

```
.\run-demo.ps1 03
```

Students complete LANGUAGE_CONTRAST.md.